

A Q-learning approach for long-term hydrothermal dispatch

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Abstract. The growing uncertainty and dimensionality of hydrothermal power systems challenge traditional stochastic dynamic programming (SDP) tools for long-term dispatch. This paper applies tabular Q-learning to a simplified Uruguayan system that aggregates four major hydro plants into one equivalent reservoir within a Gymnasium-based environment. The algorithm is trained on a three-year weekly horizon under deterministic and stochastic inflows to approximate optimal dispatch decisions without requiring an explicit transition model. Compared with MOP (the industrial SDP benchmark), Q-learning achieves virtually identical results under perfect foresight and stays within about 2.19% of cumulative cost under stochastic inflows (2,759 MUSD vs. 2,700 MUSD), while reproducing key reservoir and thermal-use patterns. These findings show that even a simple approach can deliver competitive long-term policies and provide a baseline for extending Reinforcement Learning to larger state spaces with renewable energies and storage.

Keywords: Reinforcement Learning · Q-learning · Stochastic Dynamic Programming · Hydrothermal Dispatch · Power System Optimization

1 Introduction

Electric power systems with hydrothermal generation face increasing complexity in their medium-term and long-term operation. Decisions on the weekly or monthly dispatch of hydrothermal resources must ensure reliable supply while minimizing total cost, balancing multiple technologies with distinct dynamics. In countries such as Uruguay, which depends heavily on hydropower, the variability of river inflows introduces significant stochasticity. This combination of large state spaces, uncertain transitions, and long planning horizons challenges the applicability of traditional optimization methods and motivates the exploration of alternative approaches.

The classical formulation of this problem—known as optimal long-term dispatch or hydrothermal scheduling—has historically relied on mathematical programming techniques such as Stochastic Dynamic Programming (SDP). SDP models the system as a multistage decision process and applies Bellman recursion to minimize expected cost over the planning horizon [4, 6]. However, the

curse of dimensionality limits SDP to low-dimensional representations of reservoirs and inflows. In response, Pereira and Pinto proposed Stochastic Dual Dynamic Programming (SDDP) [9], which approximates the future cost function with piecewise-linear cuts obtained from forward-backward simulations. SDDP has become the standard tool for long-term hydrothermal planning and is implemented in several industrial-grade software packages.

In Uruguay, the state-owned utility UTE has developed and maintains the software *Modelo de Operación MOP*, an open-source Java platform implementing SDP and SDDP to optimize generation scheduling [8]. This software represents the first industrial-scale application of SDDP to Uruguay’s power system and is routinely used for planning studies, procurement, and investment analysis. Another widely used tool is *SimSEE*, developed at Universidad de la República [1], which also builds on SDP principles. Both platforms model hydrology using discrete Markov chains of inflows by week of the year, enabling the incorporation of stochastic transitions estimated from over a century of historical data [5].

Although these approaches have proven effective, they rely on explicit knowledge of transition probabilities and value-function approximations that must be carefully engineered. The emergence of storage technologies, and the prospect of integrating sector-coupling resources such as hydrogen, add complexity to system dynamics and enlarge the dimensionality of the state space. These trends motivate the exploration of data-driven methods that can directly learn operating policies from interactions with the environment without requiring an explicit transition model.

Reinforcement Learning (RL) has recently emerged as a promising framework to address these challenges [11, 10]. RL methods naturally cast the operation problem as a Markov Decision Process (MDP), where the state encodes the system conditions (e.g., reservoir storage, hydrological class, time of year), actions correspond to decisions (e.g., turbine releases) and rewards capture the economic cost or benefit of those actions. By repeatedly interacting with an environment, an RL agent learns a policy that maximizes the expected cumulative reward, potentially bypassing the need for complex analytical formulations.

Among RL techniques, Q-learning stands out as a simple but powerful tabular algorithm that can serve as a benchmark for more advanced approaches. Despite its simplicity, Q-learning has been shown in other domains to approximate optimal policies for finite-state MDPs and to adapt to stochastic dynamics. Its application to hydrothermal scheduling remains limited and underexplored. Recent surveys [10, 14] highlight the potential of RL for energy systems but also emphasize the scarcity of real-world prototypes comparing RL with established SDP/SDDP-based tools.

Building upon this context, the present article applies tabular Q-learning to a simplified model of the Uruguayan power system, aggregating the four main hydro plants into a single equivalent reservoir and including thermal generation. The proposed environment is implemented in *Gymnasium* [12], allowing a modular and reproducible test bed for future RL research. Two scenarios are analyzed: (i) deterministic inflows, representing perfect hydrological fore-

casts; and (ii) stochastic inflows, representing realistic uncertainty through non-homogeneous Markov chains.

The contributions of this work are threefold:

1. The development of a Gymnasium-based environment that captures the long-term hydrothermal dispatch problem of Uruguay using over a century of historical inflow data.
2. The application and assessment of tabular Q-learning to approximate optimal dispatch policies under both deterministic and stochastic inflow conditions.
3. An empirical discussion of Q-learning’s performance relative to the established SDP baseline implemented in MOP, highlighting strengths, limitations and directions for integration with classical methods.

By explicitly comparing a Reinforcement Learning algorithm with Uruguay’s industrial-grade planning tool, this study aims to provide a benchmark and an entry point for future research on hybrid optimization–learning approaches for long-term power system operation.

Finally, the remainder of this work is organized as follows. Section 2 introduces the long-term operation of the Uruguayan power system and presents the simplified case-study model together with the related work. Section 3 details the methodological framework, including the physical model of the power system, the tabular Q-learning implementation, and the experimental setup. Section 4 defines the deterministic and stochastic scenarios used for evaluation, while Section 5 reports and discusses the results obtained by the proposed approach compared with the MOP model. Finally, Section 6 summarizes the main findings and outlines directions for future research on scalable Reinforcement Learning methods for hydrothermal dispatch.

2 Long-term operation of the power system

The long-term operation of a hydrothermal power system involves planning the optimal use of hydropower and thermal generation over extended time horizons under significant uncertainty. This section introduces the simplified Uruguayan system used as a case study and describes how its main components and constraints are represented. It also reviews related work in the literature, highlighting traditional approaches and recent advances in Reinforcement Learning applied to similar problems.

2.1 Problem description

The system considered includes low-cost and high-cost thermal plants, and a single hydro plant with reservoir and random inflows. The hydro system is modeled as an aggregate plant named *Claire*, equivalent to the combination of Bonete, Baygorria, Palmar, and half of Salto Grande. A constant energy coefficient k_{Claire}

is assumed, and the reservoir dynamics are modeled with a water-balance equation, Eq. 1 :

$$V_{t+1} = \min\left(V_t + a_t - \frac{P_t}{k_{\text{Claire}}} \Delta t, V^{\max}\right). \quad (1)$$

And the power balance Eq. 2 is:

$$P_{\text{therm,low}} + P_{\text{therm,high}} + k_{\text{Claire}} Q_t^{\text{tur}} = D_t + P_t^{\text{exp}}. \quad (2)$$

The hydrological variable v_{hidro} represents dryness/wetness and follows a non-homogeneous Markov chain with a distinct transition matrix for each week of the year. From historical inflows, 52 transition matrices of size 5×5 are built and inflows are discretized into quintiles.

The time index t is explicitly included in the state, advancing sequentially over the weekly horizon of three years (156 steps), so that seasonality in demand and inflows is properly captured.

Demand is taken as a deterministic series computed from historical hourly data. To better reflect real operation, demand is scaled by 1.2 to counteract the smoothing introduced by aggregation. Hydrological inflows are sampled from the discretized historical matrix. Exports P_t^{exp} capture any energy surplus after meeting demand, preventing infeasibilities.

2.2 Related work

The long-term scheduling of hydrothermal power systems under uncertainty has been extensively studied using Stochastic Dynamic Programming (SDP) [4, 6] and its large-scale extension, Stochastic Dual Dynamic Programming (SDDP) [9]. These approaches discretize inflows into states and optimize decisions through Bellman recursion or cutting-plane approximations. In Uruguay, UTE's MOP software [8, 13] implements SDDP and is routinely used for planning studies, while *SimSEE* [1] is widely applied in academia and consulting.

Beyond optimization-based techniques, Reinforcement Learning (RL) has gained traction as an alternative for complex energy systems [11, 10]. RL methods cast the operation problem as a Markov Decision Process (MDP) where states encode the system's conditions, actions correspond to operational decisions, and rewards represent economic outcomes. Recent studies have applied deep RL methods such as actor-critic and policy-gradient approaches to short-term dispatch, unit commitment, and storage operation [15, 3].

Tabular Q-learning, although less common in large-scale systems, remains a useful benchmark to evaluate the feasibility of RL in structured energy problems. Prior research has explored Q-learning for reservoir operation and water resource management [2], and for simplified hydrothermal scheduling models [7]. These works demonstrate that even simple algorithms can learn near-optimal policies under stochastic inflows, motivating the present study's focus on Q-learning for Uruguay's aggregated hydrothermal system.

The present article builds upon this literature by combining a century of historical inflows, a Gymnasium-based environment, and a tabular Q-learning

agent to provide one of the first systematic comparisons between RL and the established SDP/SDDP framework in a Latin American context.

3 The proposed approach

This section presents the methodological framework developed to evaluate long-term hydrothermal operation. Subsection 3.1 describes the physical model of the power system used to build the environment. Subsection 3.2 outlines the Q-learning implementation that interacts with this environment to learn operating policies.

3.1 Physical model of the power system

The modeled environment is based on a simplification of the Uruguayan electrical system, implemented in *Gymnasium*.

The focus is on the operation of a hydroelectric power plant (*Claire*), which integrates Uruguay’s four hydroelectric plants. This aggregation reduces dimensionality while preserving essential storage and discharge characteristics.

System representation. The state of the environment reflects the combined storage of the hydro reservoir, hydrological class, and time step. Hydrology uses $N_{\text{hidro}} = 5$ discrete states obtained by classifying historical inflows into quintiles (very dry to very wet). The demand is deterministic, while inflows are sampled from historical records.

Hydro plant data. Table 1 lists the main parameters of each individual hydro plant prior to aggregation (installed capacity, maximum flow, storage, and energy coefficient). Table 2 reports the resulting aggregate parameters for *Claire*. These tables document the capacities, maximum flows, and storage volumes used to calibrate the environment.

Table 1. Main parameters of individual hydro plants before aggregation.

Plant	$P^{\max}$ [MW]	$Q^{\max}$ [m ³ /s]	$V^{\max}$ [hm ³]	k [MWs/m ³]
Bonete	155	680	8200	0.23
Baygorria	108	828	0	0.13
Palmar	333	1372	1300	0.24
Salto Grande	1890	8400	3000	0.11

Model equations. Given the four hydroelectric plants in Uruguay, the *Claire* plant is modeled as the union of these.

$P^{\max}$ is the maximum power the plant can generate.

$$P_{\text{Claire}}^{\max} = P_{\text{Bonete}}^{\max} + P_{\text{Baygorria}}^{\max} + P_{\text{Palmar}}^{\max} + \frac{P_{\text{Salto}}^{\max}}{2}. \quad (3)$$

Q^{turmax} is the maximum turbine flow of the plant.

$$Q_{Claire}^{turmax} = Q_{Bonete}^{turmax} + Q_{Baygorria}^{turmax} + Q_{Palmar}^{turmax} + Q_{Salto}^{turmax}. \quad (4)$$

a_{plant} are the hydrological inflows into the reservoirs. Eq. 5 establishes how to obtain the inflows corresponding to *Claire* reservoir.

$$a_{Claire} = a_{Bonete} + a_{Baygorria} + a_{Palmar} + a_{Salto}. \quad (5)$$

For plants, a constant energy coefficient k was used.

$$k = P/Q^{tur}. \quad (6)$$

Using the Eq. 3, Eq. 4 and the data in Table 1, we derive Eq. 6, which defines the energy coefficient of *Claire*.

Table 2. Aggregate parameters of the equivalent hydro plant Claire.

Parameter	$P^{\max}$ [MW]	$Q^{\max}$ [m ³ /s]	k [MWs/m ³]	$V^{\max}$ [hm ³]
Value	1541	11280	0.137	60000

Generators. Table 3 summarizes the main parameters of the generation fleet, including installed capacities, storage volumes, and costs of thermal plants.

Where $P_{Claire}^{\max}$ represents the maximum power of the equivalent hydro plant Claire. $P_{\text{therm,low}}^{\max}$ corresponds to the installed capacity of the low-cost thermal generation unit. $P_{\text{therm,high}}^{\max}$ denotes the installed capacity of the high-cost thermal generation unit. $V_{Claire}^{\max}$ indicates the maximum storage volume of Claire's reservoir to consider the accumulation time determined in Eq. 7 and Eq. 8. The energies used on the right side of Eq. 7 correspond to the storage capacity of the corresponding real hydroelectric power plants. $C_{\text{therm,low}}$ is the variable cost of the low-cost thermal unit. $C_{\text{therm,high}}$ is the variable cost of the high-cost thermal unit.

Hydrological inflows. A long-term dataset spanning 1909–2022 is used. After aggregating the individual reservoirs into *Claire*, inflows are discretized into quintiles to define the hydrological states $v_{\text{hidro}} \in \{0, 1, 2, 3, 4\}$. Each week of the year has its own non-homogeneous 5×5 transition matrix, enabling realistic seasonal inflow patterns.

Hydro energy capacity.

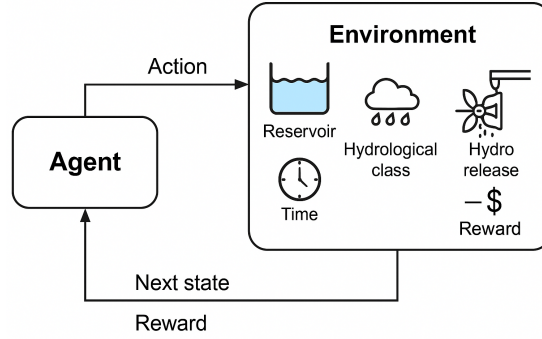
$$E_{Claire}^{\max} = E_{Bonete}^{\max} + E_{Palmar}^{\max} + \frac{1}{2} E_{Salto}^{\max} \approx 1550 \text{ GWh}. \quad (7)$$

$$\Delta t = \frac{E_{Claire}^{\max}}{P_{Claire}^{\max}} \approx 1006 \text{ hours} \approx 6 \text{ weeks}. \quad (8)$$

This capacity allows *Claire* to sustain high output for several weeks, making storage decisions critical to long-term optimization.

Table 3. Main parameters of the power system instance. Values include their respective units.

Variable	Value (with units)
$P_{\text{Claire}}^{\max}$	1541 MW
$P_{\text{therm,low}}^{\max}$	500 MW
$P_{\text{therm,high}}^{\max}$	5000 MW
$V_{\text{Claire}}^{\max}$	60000 hm ³
$C_{\text{therm,low}}$	100 USD/MWh
$C_{\text{therm,high}}$	300 USD/MWh

**Fig. 1.** Schematic diagram of the interaction between the Q-learning agent and the physical environment. The environment represents the hydrothermal power system; the agent observes the state (reservoir, hydrological class, time) and selects an action (hydro release). The environment updates the state and returns the cost-based reward.

Deterministic variables. The demand series is obtained from public MOP simulations, while hydrological inflows rely on long-term historical data. Hourly values are aggregated into weekly profiles. Demand is scaled by a factor of 1.2 to offset the smoothing introduced by aggregation.

3.2 Q-learning implementation

On top of the physical environment described above, a tabular Q-learning agent (Figure 1) is trained to approximate optimal operating policies. This subsection describes the state–action–reward structure and the training procedure, and summarizes the main hyperparameters in Table 4.

State, actions and reward. The state vector comprises three elements: reservoir volume, hydrological state, and time step. Actions represent discretized weekly turbine releases; the continuous range of feasible releases is divided into 20 levels. The reward is defined as the negative of the thermal generation cost incurred during the week, aligning the maximization of cumulative reward with cost minimization.

Q-learning update rule. The agent updates its Q-table according to Eq. 9, where s_t denotes the state at time step t , a_t the chosen action, α the learning rate, γ the discount factor, and r_t the immediate reward. Action selection follows an ϵ -greedy policy with a constant exploration rate.

$$Q(s_t, a_t) \leftarrow (1 - \alpha)Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') \right]. \quad (9)$$

Hyperparameters. Table 4 summarizes the main hyperparameters used. The Q-table size is determined by the discretization of the state and action spaces described above, resulting in a matrix of 7800×20 entries. The training is performed over multiple episodes to ensure convergence.

Table 4. Main hyperparameters of Q-learning used in the experiments.

Hyperparameter	Value
Learning rate α	0.001
Discount factor γ	0.99
Exploration rate ϵ	0.01
Episodes N_{episodes}	3000
Q-table size	7800×20

This design allows evaluating Q-learning in a transparent, interpretable setting before scaling to larger or function-approximation-based algorithms.

4 Case studies

To evaluate the proposed approach, two case studies are defined over a three-year weekly horizon (156 time steps). Both cases use the same system configuration described in Section 2, but differ in the treatment of hydrological inflows. In each case, results are obtained both with the tabular Q-learning algorithm and with the reference MOP model implementing SDP, enabling a direct comparison between a Reinforcement Learning approach and a state-of-the-art stochastic optimization tool.

4.1 Deterministic scenario

In the deterministic scenario, future hydrological inflows are assumed to be perfectly known over the 156-week horizon. This case provides a benchmark to assess whether Q-learning can converge to policies approaching the optimal dispatch derived by MOP under perfect information. Demand series are fixed according to historical profiles, while hydrological inflows follow the average weekly values estimated from the long-term dataset. No stochasticity is introduced into reservoir inflows during training or evaluation. This scenario thus tests the ability of Q-learning to exploit full predictability and minimize thermal generation costs,

as well as to replicate the policies obtained from MOP when no uncertainty is present.

4.2 Stochastic scenario

In the stochastic scenario, weekly hydrological inflows follow the non-homogeneous Markov chain derived from more than a century of historical data, producing realistic dry and wet conditions with seasonality. Each week of the 156-step horizon uses its own 5×5 transition matrix to sample inflows, ensuring that the system is exposed to random variability during training and testing. Demand remains deterministic, but the agent must infer the best long-term strategy without perfect foresight of inflows.

This case enables evaluation of Q-learning’s robustness under uncertainty compared to MOP’s SDP-based policies. By contrasting both tools in the same stochastic environment, the analysis highlights strengths and weaknesses of learning relative to model-based optimization, particularly in terms of thermal generation cost, reservoir trajectories, and policy consistency.

To ensure a fair comparison between learning and the SDP-based benchmark, the Q-learning agent is trained using the non-homogeneous Markov chain of inflows but evaluated on a large set of historical sequences. Specifically, 114 historical scenarios are constructed by sliding a three-year window over the 1909–2022 inflow record: scenario 1 reproduces the hydrology of 1909–1911, scenario 2 uses 1910–1912, and so forth, wrapping around at the end of the dataset. This evaluation protocol leverages the full historical variability, including multi-year droughts that are not adequately captured by the Markov chain, while keeping training consistent with the structure imposed by Bellman recursion in MOP. Because MOP’s stochastic dynamic programming stage also relies on a Markov representation for optimization, using the Markov chain for training and historical chronicles for out-of-sample simulation provides the most equitable basis for comparing Q-learning with the established industrial tool.

4.3 Computational platform

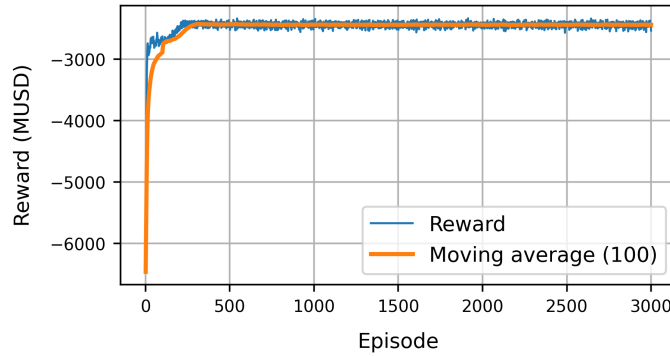
All simulations were implemented in Python using multi-threading capabilities, with the environment following the Gymnasium interface. Experiments were executed on a workstation with an Intel Core i7-8700 CPU at 3.20GHz and 16GB RAM, running Windows 11 Pro 64-bit. Computation times for each case study are reported in Section 5.

5 Results and discussion

This section reports the performance of Q-learning under deterministic and stochastic inflows. A summary of the results is presented in Table 5 where reward is represented as the opposite of the cost.

Table 5. Reward and computational time for 156-week case studies.

Case	Reward [MUSD]	Time [s]
Deterministic RL training	-2438	104
Deterministic RL evaluation	-2481	15
Stochastic RL training	-2809	396
Stochastic RL evaluation	-2759	41
Deterministic MOP optimization	–	24
Deterministic MOP simulation	-2477	0.2
Stochastic MOP optimization	–	31
Stochastic MOP simulation	-2700	97

**Fig. 2.** Q-learning training (deterministic inflows).

5.1 Deterministic inflows

In the deterministic inflows case, the agent’s reward converged during training to approximately -2438 MUSD in 1.74 minutes, and the evaluation of the unique deterministic scenario executed in 15 seconds, as presented in Table 5. In MOP, the evaluation of the optimal policy given by the Bellman Function optimized using SDP with the same scenario shows a reward of -2477 MUSD. This small deviation is practically negligible and can be attributed primarily to the discretization of the control space inherent to the tabular Q-learning implementation, as well as minor differences in state aggregation and convergence tolerances between both methods. In Figure 2, the convergence for the deterministic case exhibits very low variance across episodes and rapidly approaches the optimal value, illustrating both the stability and efficiency of the learning process.

Furthermore, Figure 3 reveals how the deterministic policy keeps the reservoir level relatively low throughout the horizon, reflecting its ability to anticipate future inflows with perfect foresight and thus operate without uncertainty.

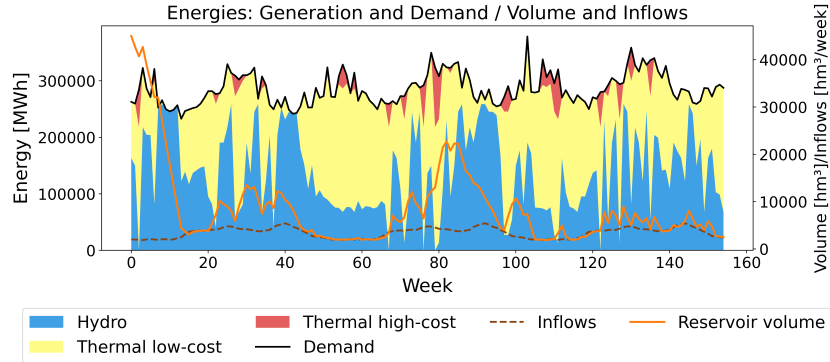


Fig. 3. Q-learning (deterministic inflows).

5.2 Stochastic inflows

In the stochastic inflows case, the Q-learning agent stabilized during training around a cumulative reward of approximately -2809 MUSD over 6.6 minutes of computation time, with the evaluation of the full set of historical scenarios executing in 41 seconds, as presented in Table 5. In MOP, the stochastic optimization stage required 31 seconds and the simulation stage 97 seconds, yielding a benchmark reward of -2700 MUSD when averaged over the same historical chronicles. This small difference between Q-learning and MOP is relatively minor given the inherent uncertainty of the problem and can be explained mainly by the discretization of the control space in the tabular approach, the stochastic nature of the learned policy, and the greater variability present in the historical scenarios used for evaluation.

Another important result observed in Figure 4 is that the convergence for the stochastic case displays higher variance across episodes and converges more slowly than in the deterministic case, reflecting the increased uncertainty in inflows and the larger effective state-action space faced by the learning agent. Furthermore, the dispatch patterns shown in Figures 5–8 illustrate how the learned policy adapts to different hydrological conditions: in dry chronicles it relies more heavily on thermal generation and maintains conservative reservoir levels, while in wet chronicles it exploits abundant inflows to increase hydraulic generation and reduce thermal costs. The averaged results over all historical scenarios confirm that, despite operating under uncertainty, the Q-learning policy tracks the main trends captured by the SDP-based benchmark and yields comparable overall costs with reasonable computational effort. In these figures, the black line is system demand, the filled areas indicate generation by source (blue: hydro; yellow: low-cost thermal; red: high-cost thermal), the dashed line shows inflows, and the orange line tracks reservoir volume. Despite spanning three years, the figures highlight the referred chronicle in the second year (steps 52–103), situated in the middle.

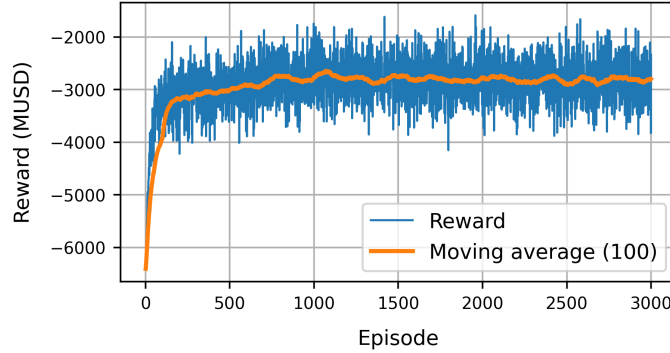


Fig. 4. Q-learning training (stochastic inflows).

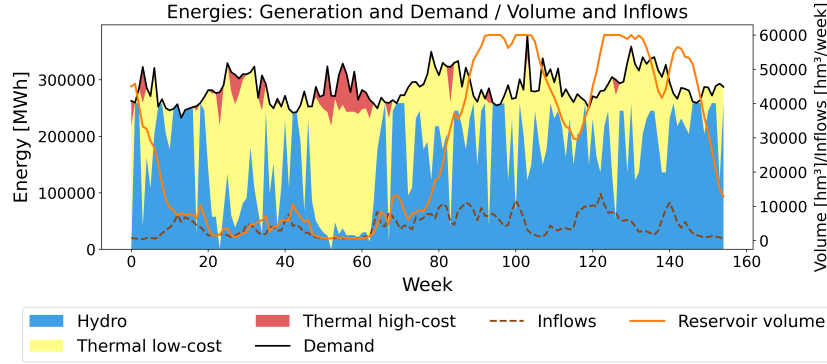


Fig. 5. Chronicles of 1913-1915 with Q-learning (stochastic inflows).

Figure 5 displays the weekly evolution of generation, demand, inflows and reservoir volume for a water-abundant chronicle. During the initial phase (approximately weeks 0–60) the reservoir remains relatively depleted and thermal generation, including high-cost units, makes a substantial contribution, which translates into elevated thermal costs. Starting around week 60, inflows increase and the reservoir recovers; hydraulic generation then rises markedly while thermal production declines, yielding a notable reduction in operational thermal costs and a smoother reservoir trajectory.

In Figure 6 throughout the horizon, thermal generation contributes persistently with both the low-cost and high-cost units, leading to generally elevated thermal costs. Occasional inflow events enable temporary increases in hydraulic generation and cost reductions, but these episodes are insufficient to sustain reservoir recovery: after each inflow peak, the reservoir volume declines rapidly again, indicating that the policy (or system constraints) does not achieve durable storage replenishment. In short, the figure illustrates the canonical drought be-

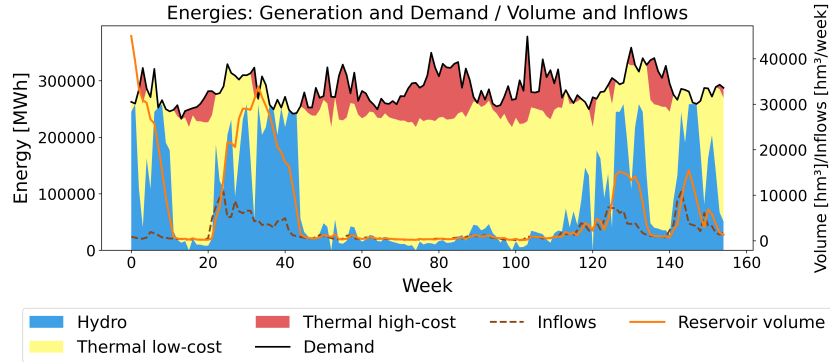


Fig. 6. Chronicles of 1916-1918 with Q-learning (stochastic inflows).

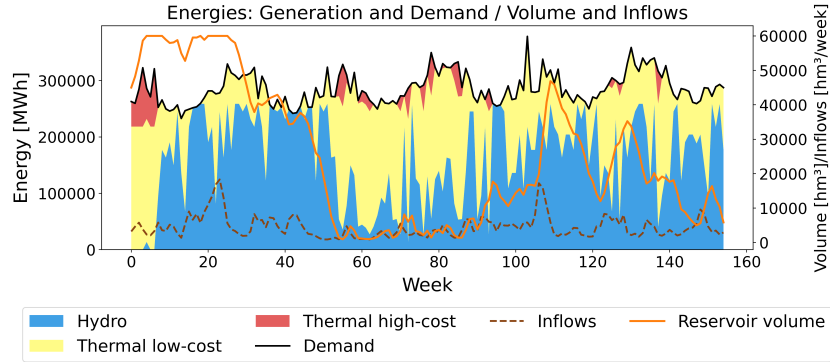


Fig. 7. Chronicles of 2017-2019 with Q-learning (stochastic inflows).

havior: continuous reliance on thermal plants, brief hydraulic substitutions following inflows, and rapid depletion of storage after those events.

Figure 7 depicts the weekly evolution of generation, demand, inflows and reservoir volume for a chronicle representing near-average hydrological conditions. In this scenario, inflows are neither persistently scarce nor excessively abundant, resulting in an intermediate operational profile. The reservoir level oscillates around a moderate band, with the agent alternating between hydraulic and thermal generation according to seasonal inflow patterns. Low-cost thermal plants complement hydropower during dry spells, while high-cost thermal units are used sparingly, leading to thermal costs that remain between those of the wet and dry cases. This figure thus illustrates the baseline behavior of the learned policy under normal conditions.

Figure 8 presents the dispatch averaged over the full set of 114 historical chronicles, offering an aggregate view of the policy’s performance under uncertainty. The averaged profiles reveal a balanced use of hydraulic and thermal resources, moderate reservoir drawdowns, and smoothed inflow patterns compared

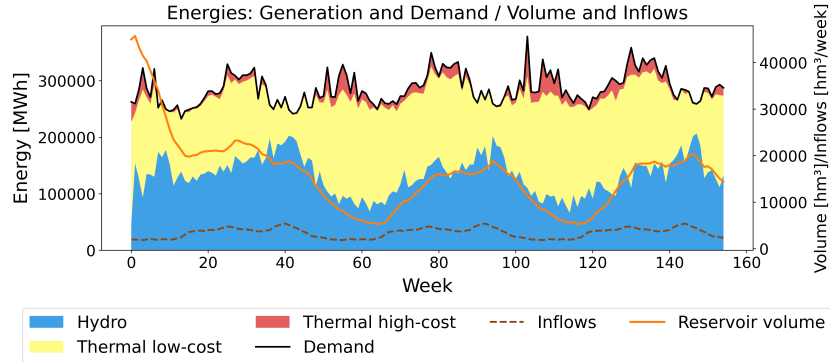


Fig. 8. Average computed over the full set of chronicles (1909–2022) with Q-learning (stochastic inflows).

with individual chronicles. Thermal generation remains significant but stable, while hydraulic output captures the general seasonal dynamics of the system. This aggregated result highlights that, despite operating under uncertainty, the Q-learning policy converges toward a robust long-term strategy whose overall cost and storage trajectories remain close to those produced by the SDP-based benchmark.

For conciseness, no equivalent figure is included for the MOP/SDP benchmark over the 114 historical chronicles, as its aggregated dispatch profiles are practically indistinguishable from those obtained with the Q-learning policy at this scale.

In short, the figures illustrate the causal chain ‘inflows $\rightarrow$ reservoir volume $\rightarrow$ hydraulic share $\rightarrow$ thermal costs’: dry periods drive reliance on thermal plants and high costs, whereas wet periods enable hydraulic substitution and cost savings.

6 Conclusions and future work

With deterministic hydrological inflows, the Q-learning agent consistently converged to a policy essentially matching the optimal dispatch derived by the SDP-based benchmark, with only negligible deviations attributable to control discretization and minor aggregation effects. Under stochastic inflows, the agent was trained on a non-homogeneous Markov chain but evaluated over 114 historical chronicles, achieving average costs within a small margin of the industrial MOP model. Although the learned policy is not optimal for every individual trajectory—reflecting the absence of perfect foresight and the increased uncertainty—it captures the main operating patterns and provides a robust approximation to the long-term optimal strategy.

This study demonstrates that tabular Q-learning can approximate long-term operation policies without an explicit transition model, offering a complement to

SDP-based tools such as MOP. Future work will extend the state space to include climatic predictors (e.g., El Niño/La Niña indicators) to reduce hydrological uncertainty, and will incorporate additional system components such as variable renewable generation and battery energy storage to capture emerging flexibility requirements. These extensions will allow testing hybrid strategies that coordinate hydro, thermal, renewable, and storage resources under uncertainty. In parallel, RL-derived results will be systematically contrasted with optimization-based tools such as MOP for objective comparison. Implementing a reduced version in MOP and benchmarking it against real system operation would help validate findings in practice-oriented scenarios and guide their eventual industrial application.

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